# REM-respiration bimodality as a candidate marker of unstable sleep states

*A single-subject longitudinal observation, prepared for the attention of Dr. Robert Thomas*

*3.5 years of consumer-wearable respiratory monitoring · 4 months post-intervention follow-up · GMM-based bimodality scoring · paired bradycardia / oscillating-respiration phenotype that resolves with the bimodality.*

## Note to Dr. Thomas

Dr. Thomas — this short technical note accompanies my covering email. It is condensed from a longer write-up prepared for a different audience; everything relating to product, commercial, or competition framing has been removed so the physiology can stand on its own.

The observation that I would most value your view on is a single-subject pattern in which paired bradycardic excursions and visibly oscillating breathing-rate traces during sleep co-occurred with a stable bimodal respiratory phenotype across 3.5 years, and then all three features resolved together over a four-month window following a non-pharmacological environmental intervention. The pattern reads — to my eye, as a non-physiologist — as something close to the unstable-state phenotype that your cardiopulmonary-coupling framework was developed to identify. I would be grateful for your judgment on whether that reading is reasonable or strained.

Specific questions for your review are listed at the end of this note.

## Summary of observations

- **Bimodal-fraction collapse, ~90% → 0%.** Across 46 baseline nights spanning three observation windows (Oct 2022, Mar 2025, Apr 2025), 86–94% of REM segments showed bimodal respiration structure on per-segment GMM scoring (Ashman's D 2.36–2.40). Post-intervention: 35% in March 2026 (partial intervention), 0% in April 2026 (full intervention).
- **Reduction in per-night in-REM respiratory variability.** Mean within-REM RMS deviation: 1.21 brpm baseline → 0.91 brpm post-intervention (cross-surface average), sustained for four months.
- **Paired bradycardic / oscillating-respiration pattern resolves alongside the bimodality.** In the pre-intervention 3.25-year window, 33% of weekly resting-heart-rate values fell into a left-tail bradycardic shoulder (≤48 bpm), and the bradycardic weekly troughs were repeatedly paired with visibly oscillating breathing-rate traces on the Garmin nightly respiration graph. Post-intervention, 0% of weekly RHR values fell into the bradycardic band over four months, and the paired oscillating-breathing pattern has not recurred.

- **Within-night temporal structure.** Bimodality severity concentrates in the 02:00–05:00 window in the n=1 record, coincident with the nocturnal ambient-temperature trough on the Highveld.
- **Stable multi-year baseline.** Three baseline periods spanning 3.5 years show bimodal fraction 86–94% and mean in-REM RMS within 0.01 brpm of each other. The intervention landed on a plateau, not a decline already in progress.

## Why I am writing to you specifically

Existing consumer-wearable autonomic metrics ask how much variability a user shows. The observation here is different in kind — it concerns which discrete state the respiratory system appears to be locked into during REM, and whether that state can be moved. The structural feature is per-segment bimodality of REM respiration: two clean Gaussian components in the within-segment respiration distribution, separated by approximately 1 brpm, weights near-equal, within-mode sigma ≈ 0.31 brpm.

Three features of the n=1 record made me think your framework might be the natural place to evaluate the finding:

- **Stable vs unstable state language.** The bimodality is not jitter around a mean; it is a structural feature that persists across years and then disappears. It reads as a sleep-state phenomenon rather than a noise property.
- **Paired bradycardia with oscillating respiration.** The pre-intervention record shows recurrent bradycardic weekly RHR troughs co-occurring with visibly oscillating breathing-rate traces. This is the pattern your CPC / cyclic-variation-in-heart-rate work was built to identify, observed here on consumer-wearable data with no overnight ECG.
- **Co-resolution.** Bimodality, bradycardic excursions, and oscillating-breathing traces all moved in the same direction at the same time in the same subject. A subject whose RHR occasionally dips into the mid-40s without paired respiratory oscillation is plausibly an athletic resting baseline; the same subject with paired oscillating breathing — and a bimodal REM respiratory structure — looks like a different phenotype.

Whether this constellation maps onto unstable-state physiology in the sense you have characterised it, or whether the bimodality has a more prosaic explanation I am missing, is the question I would most value your read on.

## The data

### Figure 1 — Bimodal-to-unimodal transition in within-REM respiration distribution

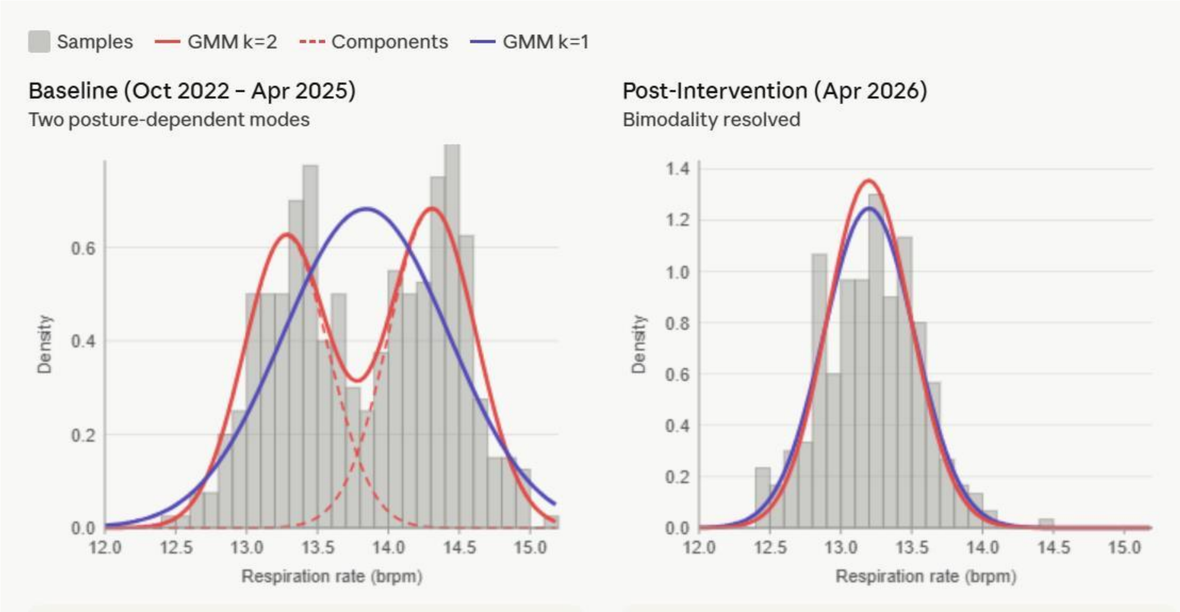


*Figure 1. Within-REM respiration-rate distribution, n=1 record. Left: baseline (Oct 2022 – Apr 2025) — two clear peaks at ≈13.3 and ≈14.3 brpm; red k=2 GMM fit tracks the bimodal structure while the purple k=1 fit visibly fails to capture the two humps; dashed red lines show individual Gaussian components. Right: post-intervention (Apr 2026) — a single peak at ≈13.2 brpm where k=1 and k=2 fits overlap completely; bimodality is absent.*

## Figure 2 — Per-night in-REM respiratory variability over four years

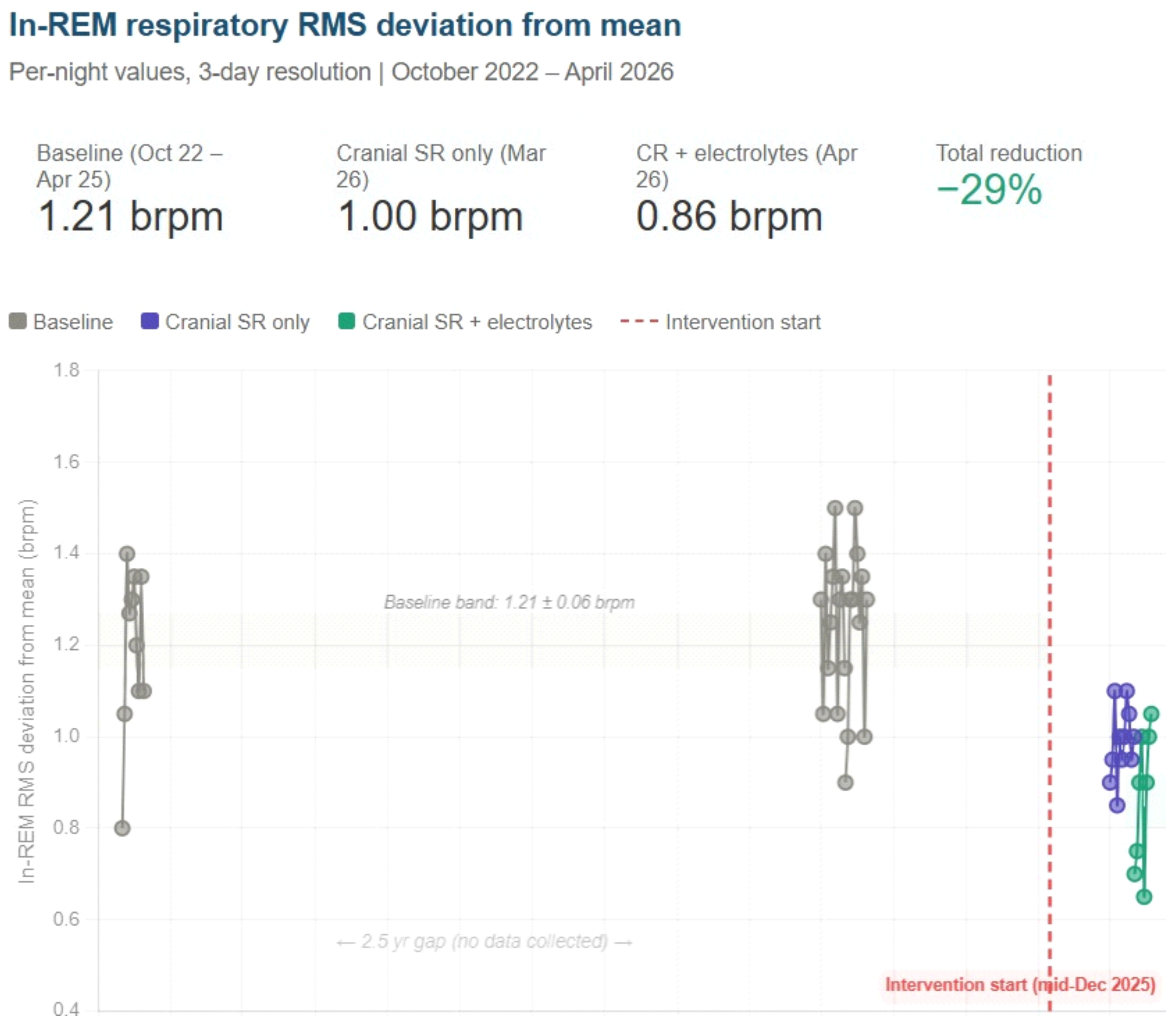


*Figure 2. Per-night in-REM RMS deviation from the within-segment mean, 3-day resolution, Oct 2022 – Apr 2026. Baseline 1.21 brpm; cranial sleep-surface intervention only (Mar 2026) 1.00 brpm; combined intervention (Apr 2026) 0.86 brpm. Intervention started mid-December 2025. The Oct 2025 – Jan 2026 gap is a true data outage (the device was on loan), not a methodological exclusion.*

The reduction is modest in absolute terms (≈0.30 brpm) but consistent across two intervention phases and stable for four months.

## Figure 3 — Within-night temporal structure of bimodality severity

Bimodality severity in the n=1 record is not uniform across the sleep period. It concentrates in the 02:00–05:00 window — the coldest hours of a typical Highveld autumn night — and is much weaker in the early-evening and post-dawn windows. The intervention effect tracks the same time-of-night pattern: the largest absolute reductions occur in the same window where baseline severity peaks.

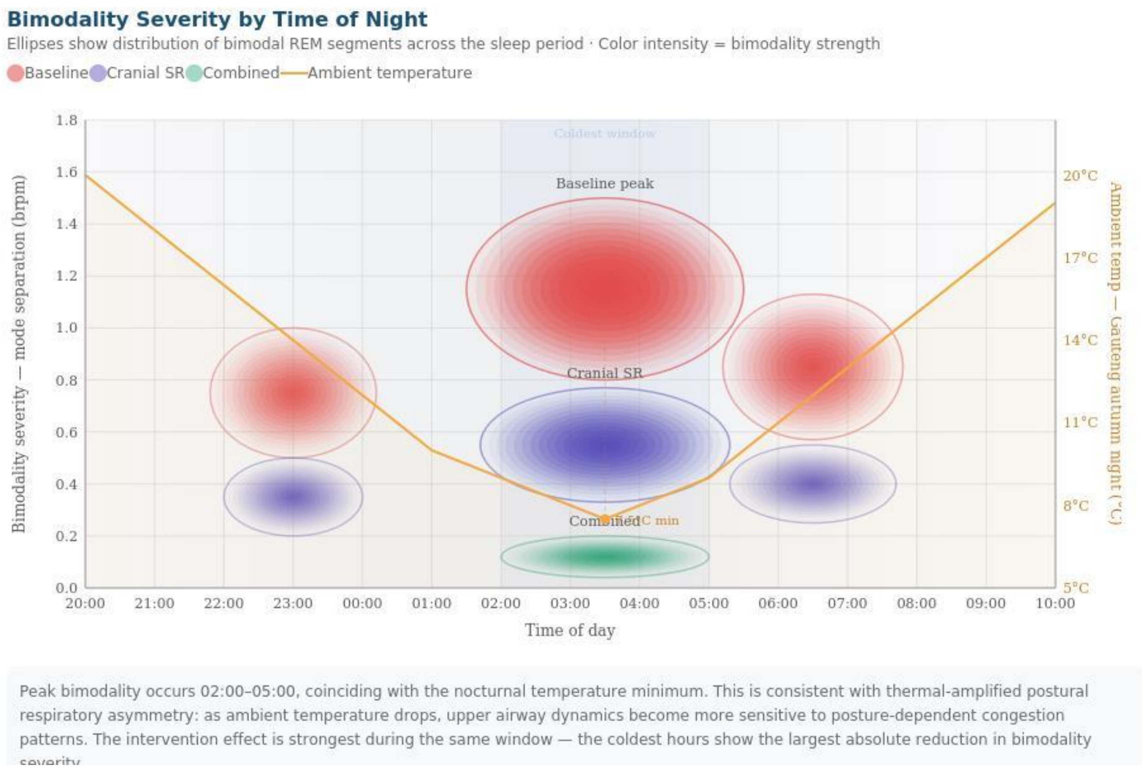


*Figure 3. Bimodality severity by time of night, n=1 record. Ellipses show the distribution of bimodal REM segments across the 20:00–10:00 sleep window for three conditions: baseline (red), cranial sleep-surface intervention only (purple), and combined intervention (green); ellipse opacity reflects bimodality strength. Orange curve: typical Gauteng autumn overnight ambient temperature profile (right axis). The temperature minimum at ≈04:00 sits directly beneath the baseline bimodality peak.*

**What this is and is not.** The figure shows a within-night temporal correlation between bimodality severity and ambient temperature. It does not, on its own, demonstrate thermal causation — the alignment is also consistent with circadian-phase effects, the late-night concentration of REM sleep itself, or ultradian autonomic cycling. I include it as a falsifiable prediction, not as a causal claim.

## Figure 4 — Weekly resting heart rate, pre- vs post-intervention

This is the figure I would most value your eye on, given that the pre-intervention bradycardic shoulder co-occurred with visibly oscillating respiration traces — the pairing that motivated the bimodality analysis in the first place.

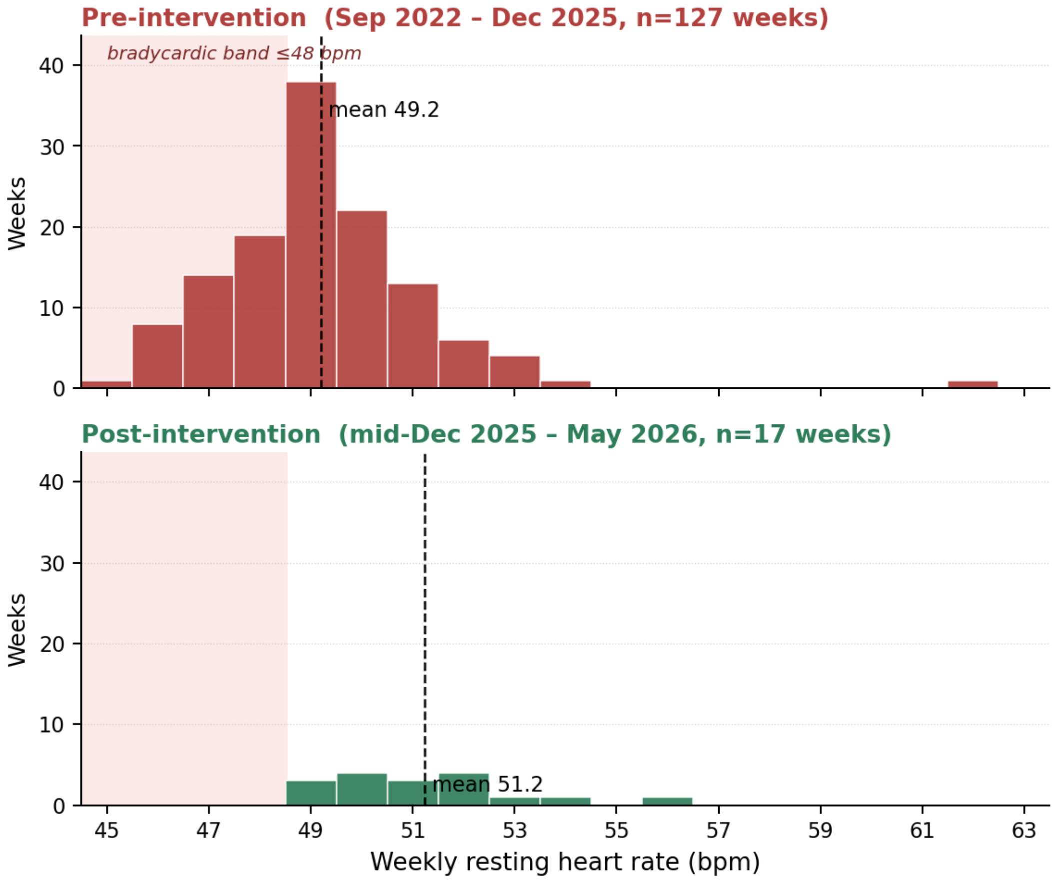


*Figure 4. Weekly resting heart rate (Garmin-computed), stacked pre/post intervention, same axes on both panels. Top (red): pre-intervention weeks, Sep 2022 – mid-Dec 2025 (n=127); mean 49.2 bpm; 33% of weeks fall in the bradycardic band ≤48 bpm. Bottom (green): post-intervention weeks, mid-Dec 2025 – May 2026 (n=17); mean 51.2 bpm; 0% of weeks fall in the bradycardic band.*

**What the figure does not show on its own.** The mean shifts by only 2.0 bpm (49.2 → 51.2). The more interpretable change is the disappearance of the left-tail bradycardic shoulder rather than the mean shift itself. The pairing with the oscillating respiratory trace is the feature that, to my reading, takes this from a mean-shift in a single subject to something that might be physiologically interpretable.

## Method — per-segment GMM bimodality scoring

Method specification, in case it is useful for assessing whether the scoring criteria are reasonable. The same pipeline would be applicable to any per-night respiration sample series with sleep-stage labels.

### Inputs

- Per-night respiration samples during REM-classified segments only (Garmin's existing on-device sleep staging).
- REM segments of duration ≥10 minutes (primary), with sensitivity analysis at ≥5 minutes.

**Per-segment analysis**

- Fit Gaussian Mixture Models with k=1 (unimodal) and k=2 (bimodal) to the within-segment respiration distribution.
- Compare via Bayesian Information Criterion. A segment is classified as bimodal if *all* of: $\Delta BIC = BIC(k=1) - BIC(k=2) > 6$; both component weights $\geq 0.25$; mean separation between components $\geq 0.5$ brpm.

**Per-night and per-user aggregation**

- **Bimodality Index (BI).** Fraction of REM segments per night classified as bimodal.
- **Within-user stability.** Standard deviation of BI across nights within a 30-day window.

**Empirical GMM parameters in the n=1 record (baseline)**

- Component means: REM_mean − 0.5 brpm and REM_mean + 0.5 brpm.
- Component σ (each): ≈ 0.31 brpm.
- Component weights: approximately equal (0.45–0.55 each).
- Ashman's D: 2.36–2.40 during baseline; 1.47 under partial intervention; undefined (single mode) post-full-intervention.

These parameters are stable across the three baseline periods to within rounding, which is one of the reasons I am taking the finding seriously enough to write.

## What makes the post-intervention reductions interpretable

Four properties of the dataset matter for whether the post-intervention numbers can be read as anything more than n=1 noise:

- **Stable baseline.** The three pre-intervention periods (Oct 2022, Mar 2025, Apr 2025) agree to within 0.01 brpm on mean in-REM RMS and to within 8 percentage points on bimodal fraction. The intervention did not land on a baseline already in motion.
- **Sustained outcome.** Four months of post-intervention stability is past the window in which placebo or novelty effects in autonomic research typically decay (≈2–6 weeks).
- **Failed washout (preliminary).** Withdrawal of the intervention in April 2026 did not produce a return to baseline variability over the available follow-up. This is consistent with a transition into a persistent state rather than a temporary forcing effect, but I treat it as preliminary; a formal extended washout has not been completed.
- **Aircon-free normalisation.** The post-intervention condition has held under higher ambient temperatures (≈26 °C, 2026) than prior baseline years (≈22 °C, with air conditioning), arguing against simple environmental cooling as the explanation.

A January 2026 transient overshoot in resting heart rate after device-use resumption — an early-adaptation phase rather than a step change — is preserved in the data and is one reason a firmware respiration-algorithm change is unlikely to explain the figures.

## Candidate mechanism (offered as a research question, not a claim)

The intervention pathway used in the n=1 record was mechanical stochastic resonance delivered through the residual mechanical signature of a low-noise circulation fan coupled into the sleep surface, combined in the later phase with evening electrolyte logging (K, Mg, Zn). I include this only because mechanism is the obvious next question; I do not claim to have established it.

The within-night thermal alignment in Figure 3 suggests an upper-airway / posture-dependent component (thermally amplified airway dynamics as ambient temperature drops, becoming more sensitive to small postural shifts during REM). A bimodal within-segment respiration distribution with ~1 brpm separation and posture-dependent switching would be consistent with that, but I would not want to argue the case strongly from a single subject.

## Within-subject decomposition by condition

Five conditions sampled across the observation window, included because the surface-only condition (Oct 2025) suggests that bedding-surface choice alone — independent of any active intervention — produces a substantial bimodality shift. I would expect this to be a familiar pattern to you from the upper-airway literature, but it surprised me to see it this clearly in n=1.

| Period | Surface | SR | Electrolytes | RMS (brpm) | Bimodal % | Reduction vs baseline |
|---|---|---|---|---|---|---|
| 3.5y baseline | Spring (cool) | — | — | 1.21 ± 0.06 | 86–94% | — |
| Oct 2025 | Foam (warm) | — | — | 1.06 | 53% | 12% RMS |
| Mar 2026 | Spring (cool) | ✓ | — | 1.00 | 35% | 17% RMS |
| Apr 2026 (a) | Foam (warm) | ✓ | ✓ | 0.87 | 0% | 28% RMS |
| Apr 2026 (b) | Spring (cool) | ✓ | ✓ | 0.94 | 23% | 22% RMS |

**Reading the table.** Surface change alone (Oct 2025) accounts for roughly half of the maximum bimodality reduction. The combination of surface, SR, and electrolytes produces full collapse on warm foam (Apr 2026 a) and near-collapse on cool spring (Apr 2026 b). The 23-percentage-point gap between the two combined-intervention conditions is genuine within-subject variation in the data; three explanations are consistent with this gap alone — a surface-by-intervention interaction, an order effect (foam came first), or measurement noise compounded by screenshot-based scoring. I would not adjudicate between them from n=1.

## Limitations

- **Single subject.** A single intervention date, a single device, no cross-subject replication, no blinding, no completed formal washout. Subject–experimenter overlap is total: I designed the instrument, ran the intervention, processed the data, and wrote the interpretation.
- **Long-COVID context.** I have personal long-COVID symptoms originating in February 2020. The baseline phenotype may or may not be representative of healthy individuals; this is exactly the question that would need cross-subject data to answer.
- **Screenshot-derived scoring.** The GMM parameters and bimodality fractions in the n=1 record were estimated from visual scoring of Garmin Connect respiration-trace screenshots, not from raw sample series. The Phase A specification above assumes access to raw samples; the n=1 numbers should be read as parameter estimates rather than exact values.
- **Data gap.** The Oct 2025 – Jan 2026 outage means the intervention onset is not directly captured in the data; only its early-adaptation phase and steady-state response are.
- **Consumer-wearable sampling.** All respiration data here is at Garmin's ~30 s sampling resolution. The within-mode σ of 0.31 brpm and ~1 brpm separation are favourable for bimodality detection at this resolution, but the underlying physiology is being observed through a coarse instrument.
- **No formal confidence intervals.** Segmented regression and bootstrap intervals are available on request but are not foregrounded; what is essentially a within-subject step change does not seem to me to be the kind of finding that confidence-interval theatre would clarify.

## Specific questions for your review

If you have time to comment on any of the following, I would be genuinely grateful. Even a sentence on any one of them would help me calibrate whether this is worth pursuing as a research question or whether I am overinterpreting a stable artefact.

1. Does a within-REM-segment respiration distribution with two ≈1-brpm-separated Gaussian components and approximately equal weights have a natural reading in your framework — for instance, as posture-dependent or state-dependent switching within a single REM episode — or does that structural pattern suggest something different to you?

2. Is the co-occurrence of a left-tail bradycardic shoulder with visibly oscillating breathing-rate traces in a non-athletic subject something you would expect to read as cyclic-variation-in-heart-rate / unstable-state physiology, and would the co-resolution of all three features (bimodality, bradycardic shoulder, oscillating respiration) over four months be consistent with a transition between your stable and unstable sleep regimes?
3. Is the within-night concentration of bimodality severity in the 02:00–05:00 window (Figure 3) consistent with a thermally amplified upper-airway / postural mechanism, or do you read it more naturally as a circadian or REM-architecture artefact given REM's late-night concentration?
4. If you were designing the smallest meaningful replication study, what would it look like? My current intuition is a ≈100-user × 10-night population characterisation on existing consumer-wearable data with sleep staging, with the GMM scoring run against synthetic unimodal controls for false-positive calibration. I would value your view on whether that is the right first step or whether something more physiologically grounded (e.g. a small cohort with simultaneous PSG and consumer-wearable respiration) would be a better first move.
5. Is there a body of work on bimodal-vs-unimodal within-REM respiratory distributions that I have failed to find, and if so where would you suggest I look? The framing in terms of state membership rather than variability magnitude is what I have not been able to place in the literature, and I would rather discover that someone has already done this than continue under the impression that it is novel.

## Closing note

The longitudinal record underlying this note was not collected for the bimodality analysis. It was collected from 2022 onward as part of a personal interest in autonomic-environmental interactions on the Highveld, in the context of long-COVID symptoms persisting since February 2020. The bimodality finding emerged from re-analysis of that existing dataset, which I take to be a methodological mild positive — the data was not collected with the present hypothesis in mind, reducing one common source of bias.

If Phase A on a cross-subject sample finds no above-floor population of bimodal users, the result remains a personal observation and I will treat it as such. If it finds a non-trivial subset, I would want the next step to be physiologically grounded rather than algorithmically driven, which is the main reason I am writing to you rather than only to engineering groups.

Thank you for any time you are able to give this. I am of course available to share the underlying data, the n=1 raw exports where I have them, the synthetic validation datasets, or any of the analysis code; please let me know what would be most useful.

With thanks and respect,

*[Author name]*

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